
Bridging Risk Parity and Variance Optimization: A Schur Complement Approach to Recursive Asset Allocation

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1. Motivation: The Portfolio Manager’s Dilemma

An investor ultimately strives to achieve extraordinary results. While “extraordinary” can mean different things, the goal for an investor usually involves achieving *above-average returns with average risk*, or *average returns with below-average risk* (Marks, 2011).

In this work, we propose a tool for portfolio managers that focuses on *risk minimization* as the primary objective, while seeking to preserve returns in the process.

In practice, portfolio management often oscillates between the pursuit of theoretical optimality and the need for robustness in uncertain markets. Our approach aims to make that trade-off explicit and manageable.

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Before introducing the Schur approach to portfolio construction, we briefly revisit some well-known concepts from portfolio theory, focusing on their *strengths and weaknesses*, as these form the foundation of our approach.

1.1. Mean–Variance Optimization

Mean–Variance Optimization (MVO) (Markowitz, 1952) is probably known as the most famous portfolio construction method, being at the same time equally well-known for its shortcomings. MVO, which computes portfolio weights from the inverse covariance matrix and expected returns, is notoriously sensitive to estimation error.

Despite its fragility, MVO has important strengths. It seeks a *global optimum* (given the input data) and makes full use of the available information to find it.

$$\Sigma = \begin{bmatrix} \mathbf{a} & b \\ c & \mathbf{d} \end{bmatrix} \quad (1)$$

1.2. Hierarchical Risk Parity

At the other end of the spectrum lies *Hierarchical Risk Parity (HRP)* — an elegant and influential contribution to portfolio theory, originally introduced by Mantegna (1999) and popularized by López de Prado (2016). HRP’s main advantage is its *robustness to noise*: by employing a hierarchical clustering approach, it avoids matrix inversion altogether, making it far less sensitive to estimation errors.

However, HRP also has limitations. In particular, it tends to lose *cross-cluster information*, which makes the resulting allocation potentially incomplete or too conservative (Fig. 1).

	Cluster 1	Cluster 2
Cluster 1	A	B cross-cluster
Cluster 2	C cross-cluster	D

Fig. 1. Block-structured covariance matrix under HRP.

Therefore, we don't need a replacement for these methods, but a *controlled bridge* between them — a way to “budget” how much cross-cluster information we reintroduce while maintaining robustness.

2. Schur: The Missing Bridge

Our first key observation is that MVO and HRP, while seemingly different, share deeper structural similarities. In fact, *the weights for the Minimum Variance Portfolio can be computed using a hierarchical scheme* by applying the Schur complement formula for block-matrix inversion.

$$\Sigma^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} (A - BD^{-1}C)^{-1} & 0 \\ -D^{-1}C(A - BD^{-1}C) & D^{-1} \end{pmatrix} \begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}. \quad (2)$$

As explained in [Microprediction \(2022\)](#) and later in [Cotton \(2024\)](#), augmenting hierarchical blocks with the Schur complement in the recursive bisection step of HRP makes it possible to reproduce the Minimum Variance Portfolio. In that respect, the only difference between HRP and MVO becomes the parameter $\gamma \in [0,1]$.

$$A^c(\gamma) = A - \gamma BD^{-1}C \quad (3)$$

$$b_A(\gamma) = \vec{1} - \gamma BD^{-1}\vec{1}, \quad (4)$$

$$w(\Sigma) \propto \begin{pmatrix} \frac{1}{\nu((A^c)^{*}b_A)} & w\left(\frac{A^c}{b_A b_A^T}\right) \\ \frac{1}{\nu((D^c)^{*}b_D)} & w\left(\frac{D^c}{b_D b_D^T}\right) \end{pmatrix}. \quad (5)$$

Empirical analysis shows that introducing a moderate amount of cross-asset information ($\gamma > 0$) can lower portfolio variance relative to pure HRP. Moving slightly toward HRP's robustness from MVO can improve numerical stability in ill-conditioned covariance matrices, making MVO more practical.

Building on that insight, we introduce another parameter, $\delta \in [0,1]$, for finer control. Empirical testing suggests that setting $\gamma = \delta$ does not necessarily yield the lowest portfolio variance. However, distinguishing these two parameters offers greater flexibility for investors (Fig. 2).

$$A^c(\gamma) = A - \gamma BD^{-1}C, \quad (6)$$

$$b_A(\delta) = \vec{1} - \delta BD^{-1}\vec{1} \quad (7)$$

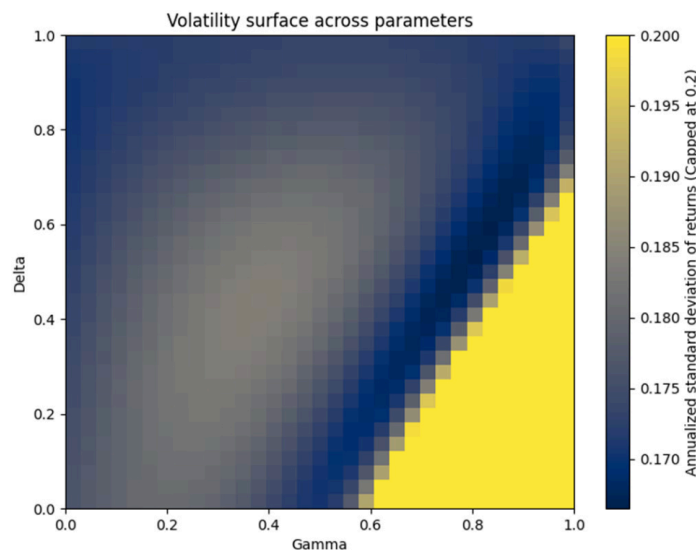


Fig. 2. Variance landscape across the parameter space.

The parameter γ controls the structural cross-cluster interaction between blocks (via Schur complements AC and DC), while δ determines how much of that information is used to tilt allocations through the b -vectors. This separation between “*admit information*” and “*use it to tilt weights*” provides a practical framework for balancing stability and efficiency.

In summary, MVO and HRP lie on a *continuum parameterized by* (γ, δ) . This perspective opens an entirely new universe of strategies that exist between the two. Conceptually, this view reframes portfolio construction as a continuum rather than a choice, a spectrum where investors can navigate between structure and noise.

3. Dynamic Parameters for Active Portfolio Management

Although this framework yields an infinite number of potential strategies, the key question in the risk management and investing world remains: *which one should be used?* In today’s ever-changing markets, relying on a single static strategy is unlikely to be sufficient.

To address this issue, we define *dynamic parameter selection* — $(\gamma(t), \delta(t))$ — guided by ongoing diagnostics. The motivation is simple: there is no free lunch, and no single strategy fits all market conditions.

Our empirical tests demonstrate that dynamically adjusting these parameters to reflect current market views and conditions can deliver significant added value (Fig. 3).

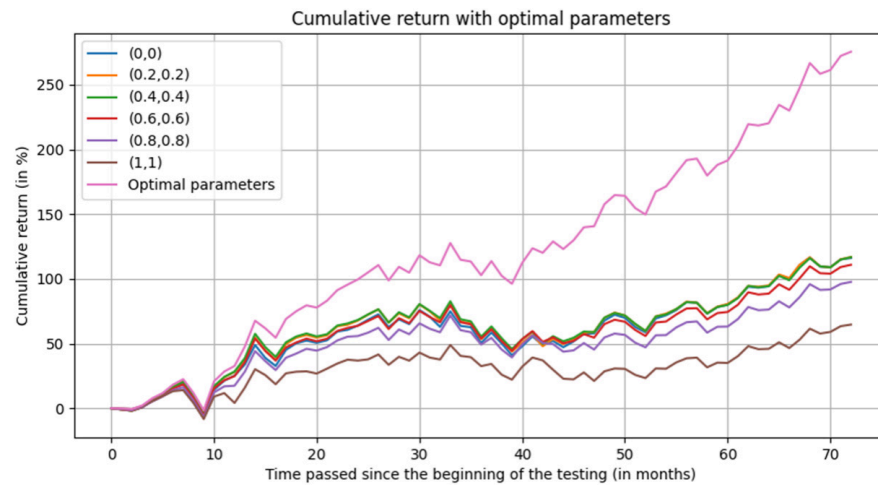


Fig. 3. Performance improvement from optimal parameter selection.

With this small shift, we change the entire narrative — from a binary choice of “the best strategy” to an *optimization and forecasting problem*. Now, the parameters $(\gamma(t), \delta(t))$ become target variables, and a fund manager who can predict them accurately gains a competitive edge.

This perspective opens the door to integrating predictive modeling and machine learning, allowing managers to forecast and adapt the allocation structure itself.

4. Conclusion

Schur-based allocation unifies two distinct portfolio construction methods into a *dynamic, information-budgeting framework*. It offers a new, practical methodology for portfolio design with clear benefits for investors.

It empowers portfolio managers to express their market views flexibly, choosing anywhere along the continuum between the stability and robustness of HRP, the informational completeness of MVO, and everything in between.

Looking forward, this framework invites a new research direction. Not the search for the “best” model, but the design of adaptive mechanisms that let investors move intelligently along the continuum.

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